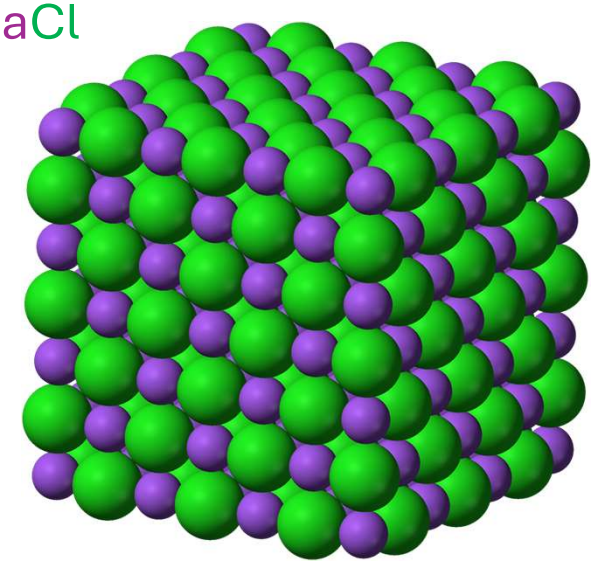
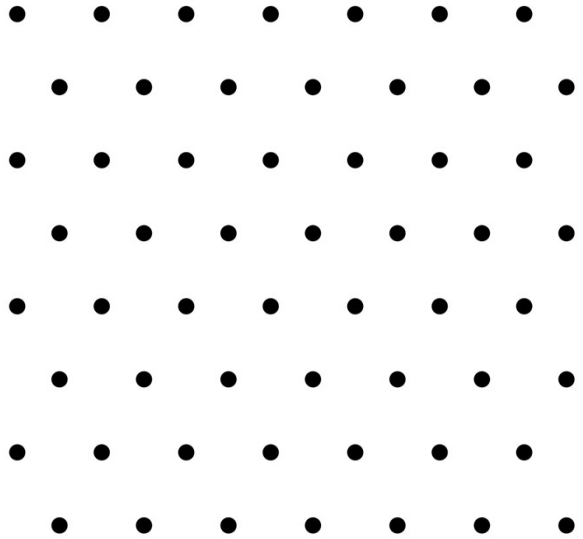
NaCl





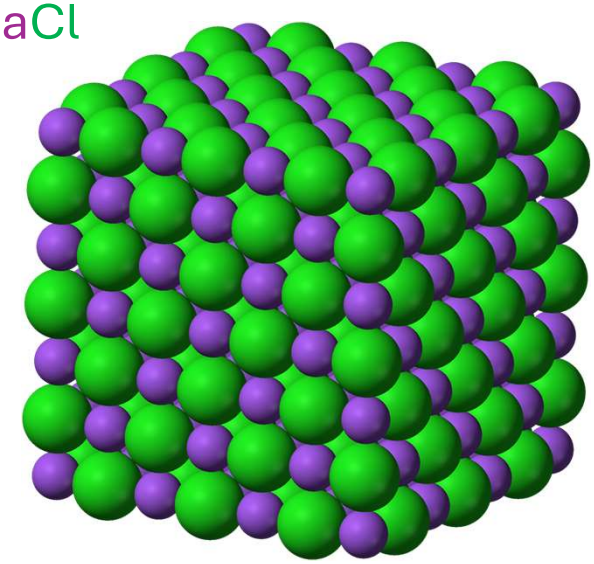
Crystal: periodic arrangement of atoms

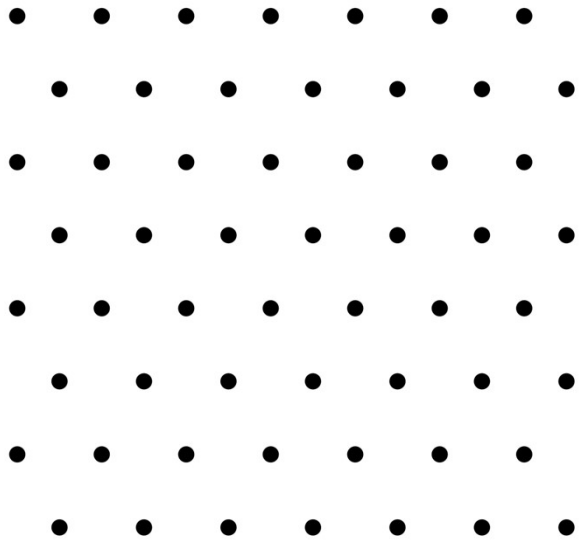
(solid phase possesing discrete translational symmetry)

Lattice: infinite set of points defined by integer sums of a set of linearly independent primitve lattice vectors (PLV).

Lattice (Def. 2): infinite set of vectors where adding of any two vactors in the set gives a third vector. (Set closed for addition of vectors.)

NaCl



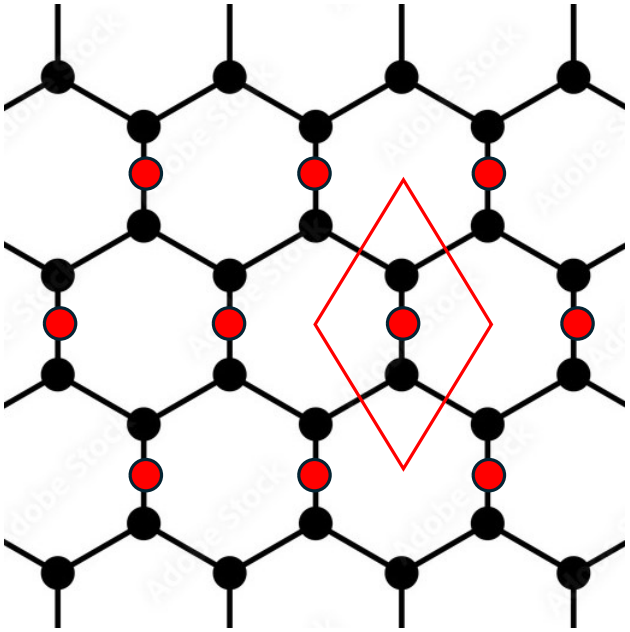


Crystal: periodic arrangement of atoms

(solid phase possesing discrete translational symmetry)

Lattice: infinite set of points defined by integer sums of a set of linearly independent primitve lattice vectors (PLV).

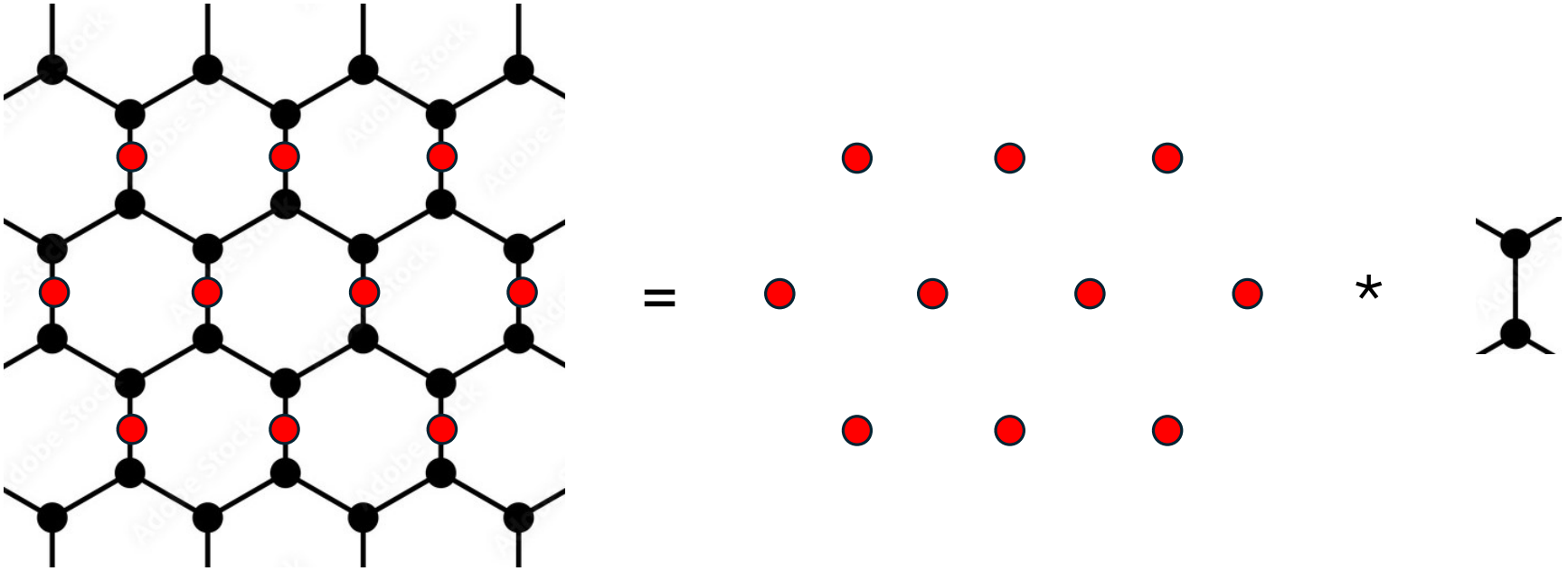
Lattice (Def. 3): Set of points where the environment of any given point is equivalent to the environment of any other given point.



Unit cell: region of space such that when many identical unit cells are stucked together it tiles (completely fills) all of space and reconstruct the full structure

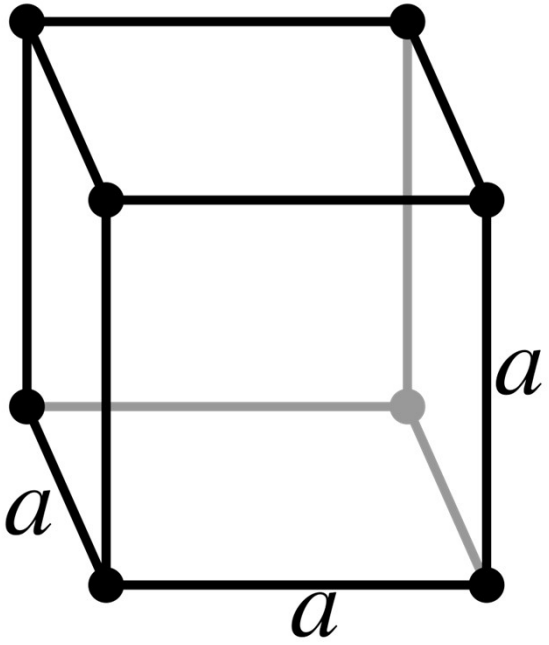
Unit cell (Def. 2): the repeated motif which is the elementary building block of the periodic structure

Primitive unit cell: smallest unit cell; contains exactly one lattice point



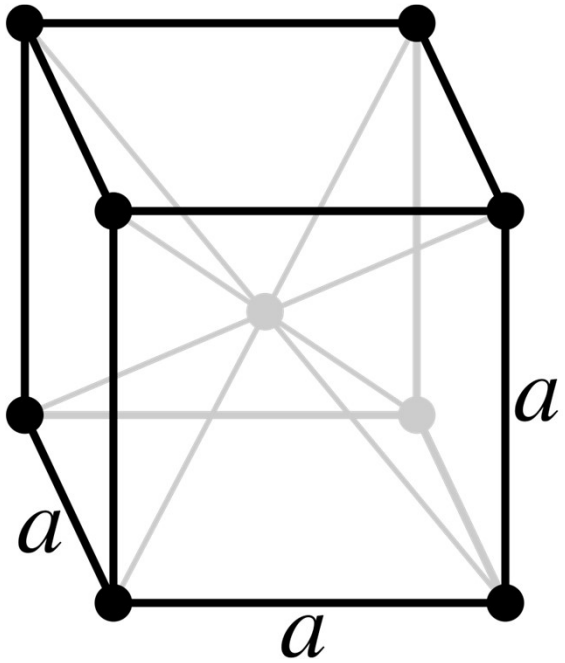
Crystal = Lattice + Basis

Simple cubic
P cubic



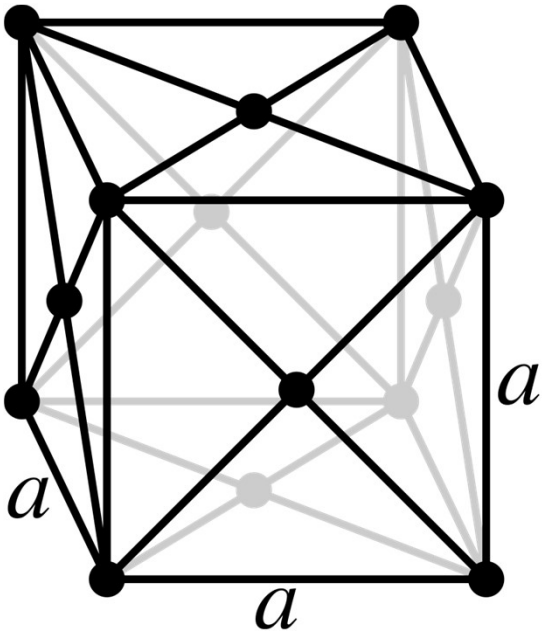
Polonium

Body centered cubic – BCC
I cubic



Li, Na, K, Fe, Mo, Cs

Face centered cubic – FCC
F cubic

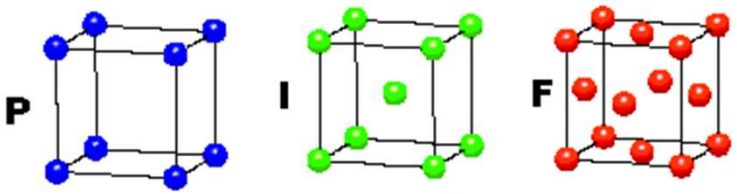


Al, Ca, Au, Pb, Ni, Cu, Ag

CUBIC

$a = b = c$

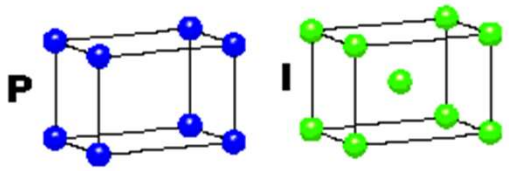
$\alpha = \beta = \gamma = 90°$



TETRAGONAL

$a = b \neq c$

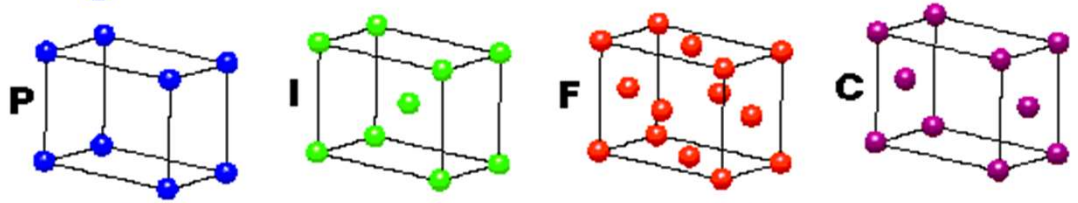
$\alpha = \beta = \gamma = 90°$



ORTHORHOMBIC

$a \neq b \neq c$

$\alpha = \beta = \gamma = 90°$

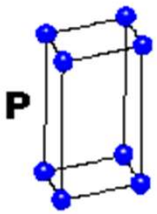


HEXAGONAL

$a = b \neq c$

$\alpha = \beta = 90°$

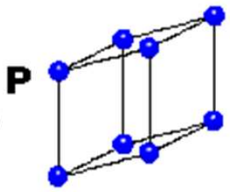
$\gamma = 120°$



TRIGONAL

$a = b = c$

$\alpha = \beta = \gamma \neq 90°$

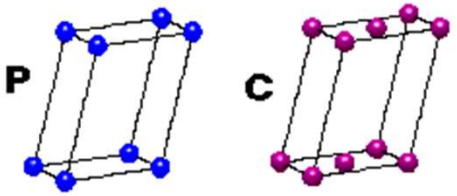


MONOCLINIC

$a \neq b \neq c$

$\alpha = \gamma = 90°$

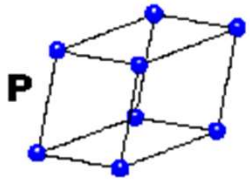
$\beta \neq 120°$



TRICLINIC

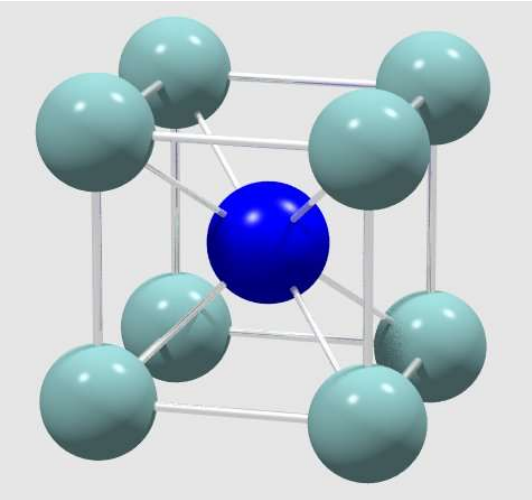
$a \neq b \neq c$

$\alpha \neq \beta \neq \gamma \neq 90°$

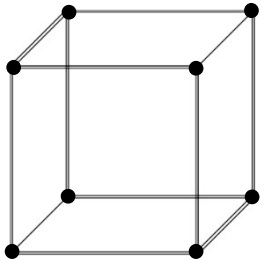


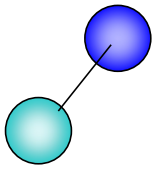
4 Types of Unit Cell
P = Primitive
I = Body-Centred
F = Face-Centred
C = Side-Centred
+
7 Crystal Classes
→ 14 Bravais Lattices

CsCl structure



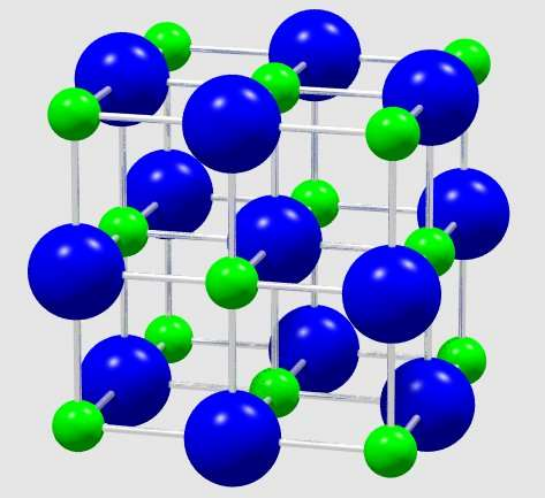
Simple cubic lattice



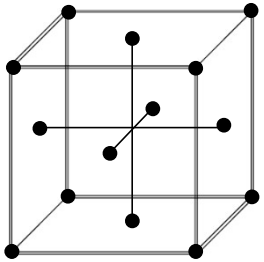


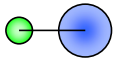
Basis:
1/2,1/2,1/2
0,0,0

NaCl structure - Rock salt



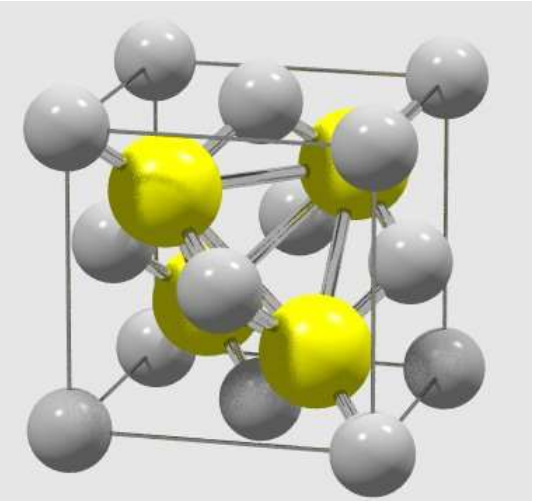
Face centered cubic (FCC)

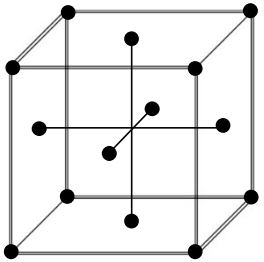


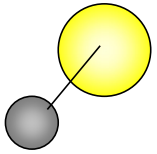


Basis:
1/2,0,0
0,0,0

ZnS structure - Sphalerite



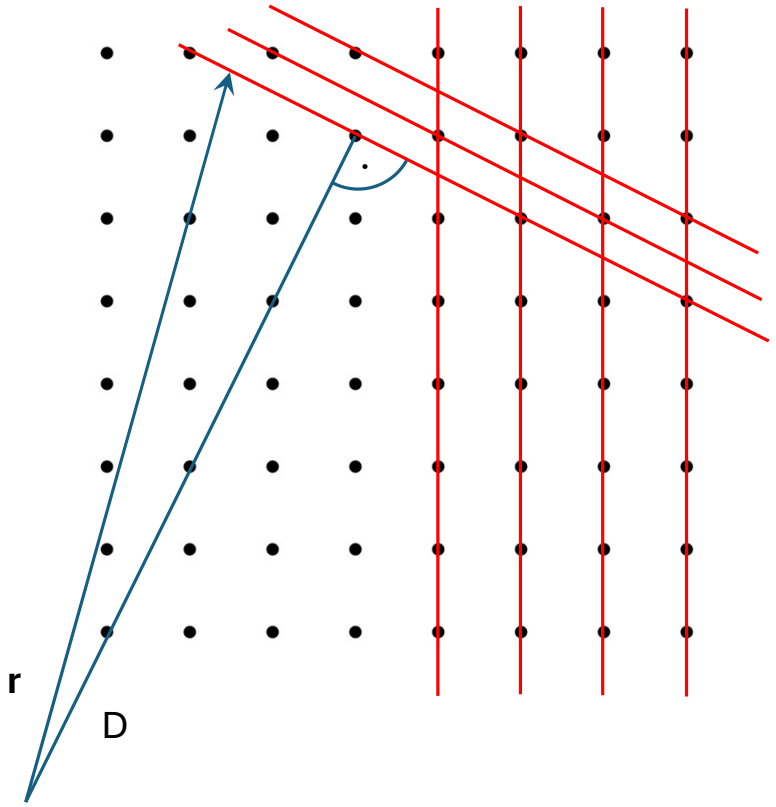




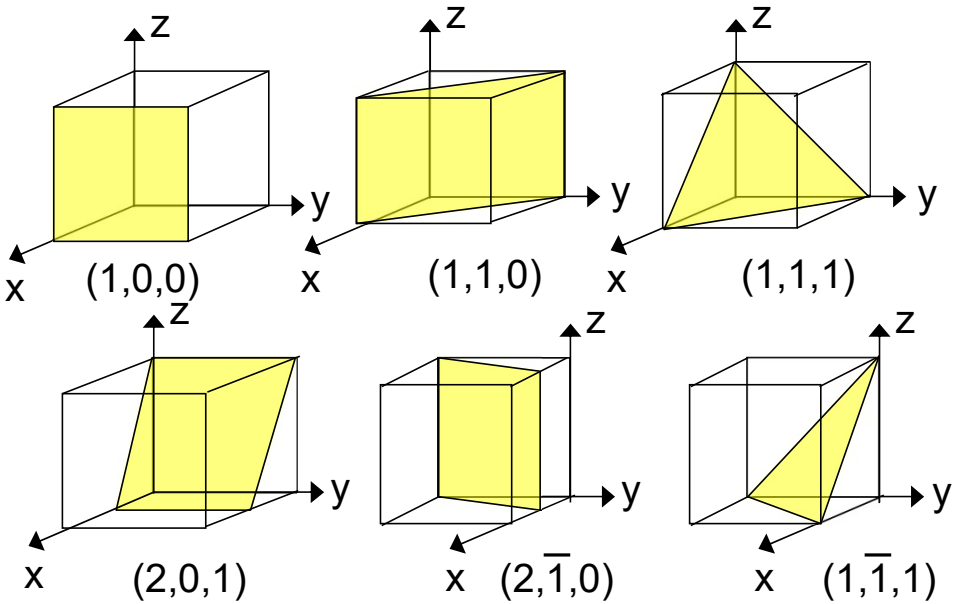
Basis:
1/4,1/4,1/4
0,0,0

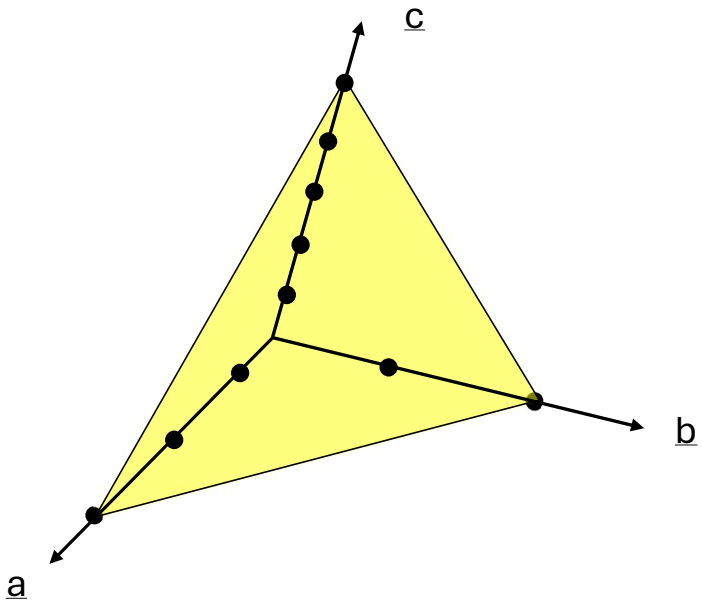
Lattice plane (crystal plane): a plane containing at least three non-collinear points of a lattice

Family of lattice planes: an infinite set of equally spaced parallel lattice planes which taken together contain all points of the lattice



Construction of **Miller-index (h, k, l)**
- Intersection of the plane with the three coordinate axes in units of the three lattice constants: 3, 2, 5
- Reciprocal of the intersects: 1/3, 1/2, 1/5
- Smallest integers with the same ratio: 10, 15, 6





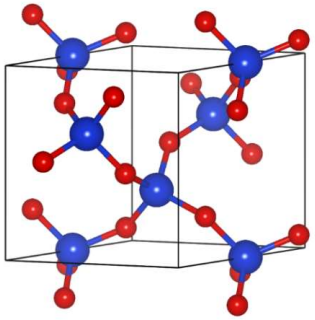
Notion of a plane (h, k, l)

Notion of a direction $[h, k, l]$

Miller-index components could be negativ $(1,\bar{1},\bar{1})$

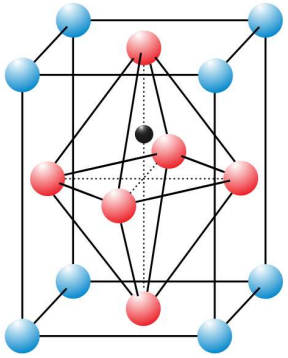
Set of symmetry equivalent planes $\{h, k, l\}$

Quartz



SiO_2

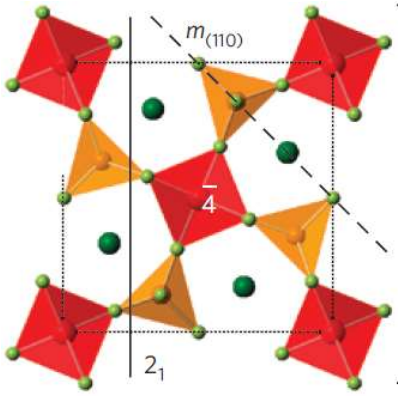
PZT: lead zirconate titanate



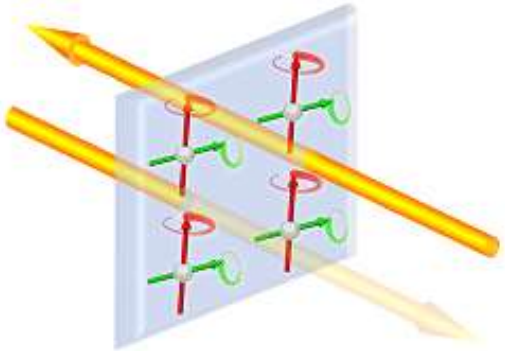
Pb^{2+} O^{2-} Ti^{4+}; Zr^{4+}

$$\left.\begin{aligned} \underline{\boldsymbol{P}} &= \underline{\underline{\boldsymbol{\alpha}}}\,\underline{\boldsymbol{H}} \\ \underline{\boldsymbol{M}} &= \underline{\underline{\boldsymbol{\alpha}}}'\,\underline{\boldsymbol{E}} \end{aligned}\right\} \quad \underline{\underline{\boldsymbol{\alpha}}} = -\underline{\underline{\boldsymbol{\alpha}}} = 0$$

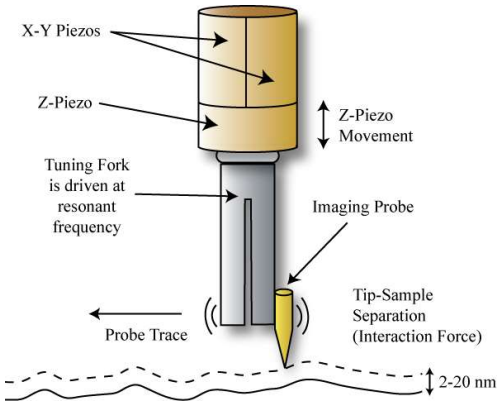
$Ba_2CoGe_2O_7$

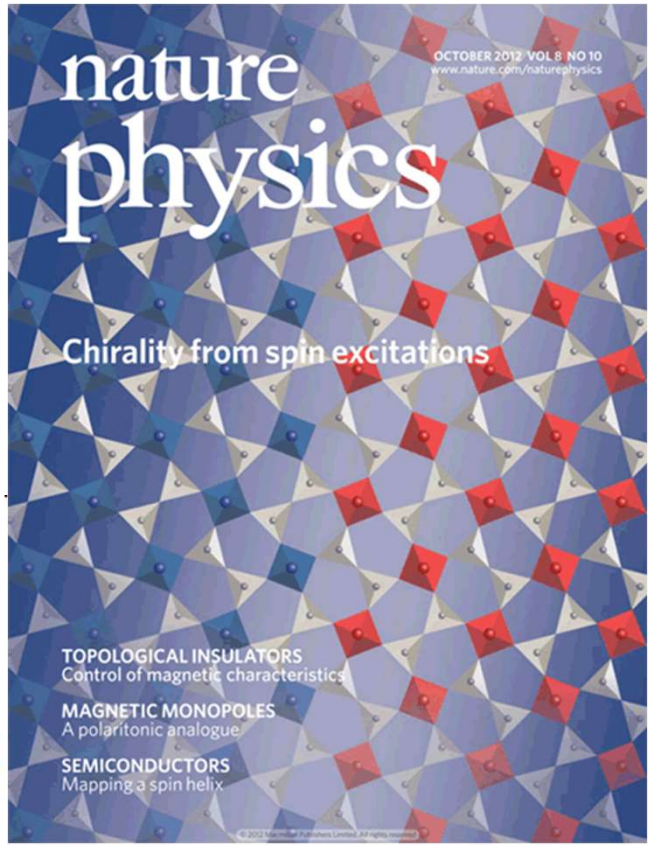


$$\underline{\boldsymbol{P}} = \underline{\underline{\boldsymbol{\alpha}}}\,\underline{\boldsymbol{H}}$$



Atomic Force Microscope (AFM)
Scanning Tunneling Microscope (STM)





S. Bordács *et al.*, Nature 26 August 2012.
I. Kézsmárki et al., Nature 6 January 2014.